

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime b

1. System Requirements

System Name: ResuFit: Precision CV Matching System

Architecture Style: Layered (N-Tier)

Functional Requirements

- Submit CV
- View and Edit CV
- Answer Screening Questions
- Get Rating for CV

Non-Functional Requirements

- Performance

Description: The system **MUST** respond to user input within 2 seconds.

- Reliability

Description: The system **SHALL** be available 99.9% of the time.

- Security

Description: The system **MUST** protect user data and **MUST NOT** store sensitive information.

- Privacy

Description: The system **MUST NOT** collect or store user data without consent.

- Scalability

Description: The system **SHOULD** be able to handle 1000 concurrent users.

- Usability

Description: The system **MUST** be user-friendly and **MUST** provide clear instructions.

- Maintainability

Description: The system **SHOULD** be easy to modify and update.

- Compatibility

Description: The system MUST be compatible with major browsers and MUST NOT have any known bugs.

- Integration

Description: The system MUST integrate with HR module from Odoo ERP software.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, co

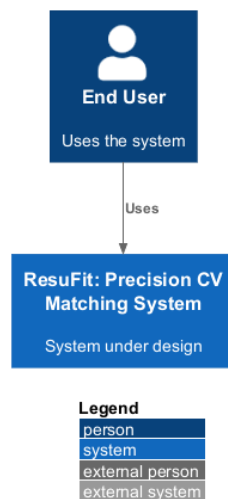
```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Layered (N-Tier)";
  Property source : String = "AI";
  Component CVParser = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Parse and analyze submitted CVs";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component CVEditor = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Allow applicants to view and edit their parsed CVs";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ScreeningQuestionManager = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Manage and present screening questions to applicants";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component RatingGenerator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Generate a percentage-based result for each applicant's CV";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Attachment CVParser.out to CVEditor.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
  }
  Attachment CVEditor.out to ScreeningQuestionManager.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment ScreeningQuestionManager.out to RatingGenerator.in {
```

```

    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CVParser.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry";
Property delivery : String = "at-most-once";
Property ordering : Boolean = true;
Property authentication : String = "centralized";
Property authorization : String = "RBAC";
}

```

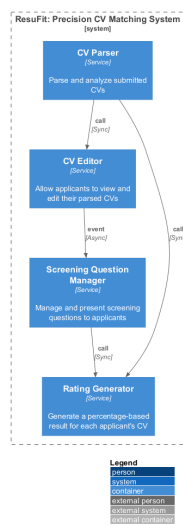
3. Context View



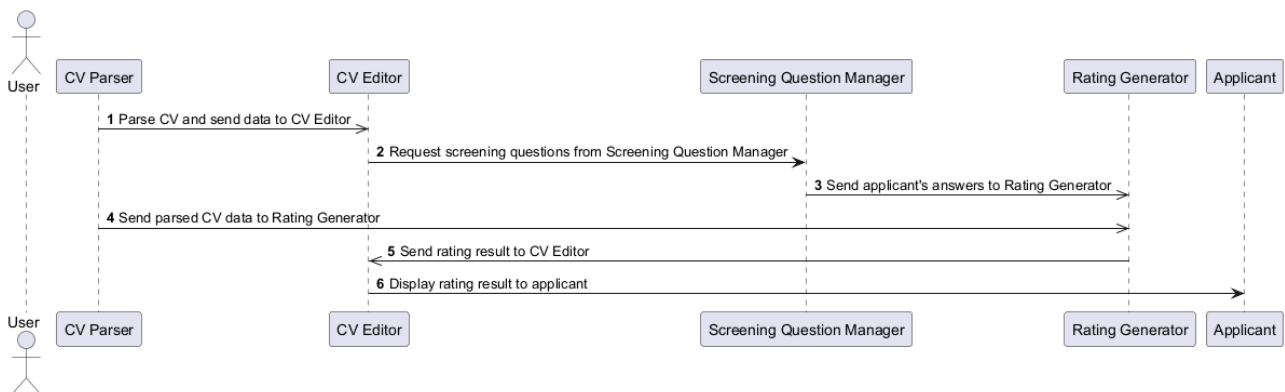
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

